



Clinical value of ^{123}I -metaiodobenzylguanidine single-photon emission computed tomography/computed tomography in spontaneous regression of localized congenital neuroblastoma: an evaluative study

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Background: Congenital neuroblastoma represents an early-onset form of neuroblastoma. This study aims to evaluate the imaging features of ^{123}I -metaiodobenzylguanidine (^{123}I -MIBG) single-photon emission computed tomography (SPECT)/computed tomography (CT) and their correlation with the clinical characteristics of localized congenital neuroblastoma.

Methods: We retrospectively collected imaging and clinical data from congenital neuroblastoma patients who underwent ^{123}I -MIBG SPECT/CT. The volumes of interest of primary lesion, muscle, lung and mediastinum blood pool were delineated by 3DSlicer. The ratios of the maximum and average counts between the tumor and muscle/lung/mediastinum blood pool were calculated to explore the factors related to the spontaneous regression of the tumor.

Results: A total of 11 patients with localized congenital neuroblastoma were included in this study. The majority of the lesions were identified prenatally. ^{123}I -MIBG imaging demonstrated that all of the spontaneous regressed lesions were localized neuroblastoma, which did not cross the midline and were negative for image-defined risk factors, with a maximum diameter of less than 3.1 cm. The ^{123}I -MIBG uptake in the tumor was significantly associated with spontaneous regression of neuroblastoma. Lower ^{123}I -MIBG uptake in the tumor is associated with spontaneous tumor regression of neuroblastoma.

Conclusions: ^{123}I -MIBG SPECT/CT is a valuable tool for the diagnosis, staging and monitoring of treatment response in congenital neuroblastoma. The ^{123}I -MIBG uptake of the tumor can predict the spontaneous regression in localized congenital neuroblastoma.

Keywords: Congenital neuroblastoma; single-photon emission computed tomography (SPECT); ^{123}I -metaiodobenzylguanidine (^{123}I -MIBG); radionuclide imaging

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Introduction

Neuroblastoma (NB) is the most common extracranial solid tumor in children, originating from immature neural crest cells. It can arise in any part of the sympathetic nervous system and accounts for more than 7% of all malignant tumors diagnosed in children under the age of 15 years old (1).

Congenital neuroblastoma, a specific type of neuroblastoma defined as early-onset neuroblastoma diagnosed during the fetal or neonatal period, accounts for approximately 5% of all neuroblastomas diagnosed annually (2). The biological characteristics and prognosis of congenital neuroblastoma differ significantly from those observed in older children with neuroblastoma. Spontaneous regression is more common in congenital neuroblastoma. However, due to the heterogeneity of NB, there are currently no universally accepted predictive biomarkers for spontaneous regression of congenital neuroblastoma (3-5).

As a specific imaging agent for neuroblastoma, ^{123}I -Metaiodobenzylguanidine (MIBG) plays a crucial role in the diagnosis, detection of metastatic lesions, and monitoring treatment response (6). Currently, there is a lack of studies evaluating the predictive value of ^{123}I -MIBG single-photon emission computed tomography/computed tomography (SPECT/CT) in spontaneous regression of the localized congenital neuroblastoma. Therefore, this study aims to evaluate the predictive value of ^{123}I -MIBG SPECT/CT in spontaneous regression of the localized congenital neuroblastoma.

Methods

Patients

We retrospectively collected the imaging data and clinical data of congenital neuroblastoma patients who underwent ^{123}I -MIBG SPECT/CT at the Beijing Friendship Hospital from January 2018 to January 2024. The inclusion criteria were as follows: (I) initial identification of the mass during the fetal or neonatal stages; (II) pathologically or clinically confirmation of NB; (III) complete clinical data and follow-up information. Exclusion criteria included: (I) mass first identified over 1 month after birth; (II) received any treatment before ^{123}I -MIBG SPECT/CT; (III) the main tumor received pathological biopsy before ^{123}I -MIBG SPECT/CT; (IV) the distant metastases were identified during ^{123}I -MIBG SPECT/CT imaging; (V) lost to follow-up; (VI) at the time of ^{123}I -MIBG SPECT/CT imaging,

the tumor had undergone spontaneous regression; (VII) termination of observation for any other reasons (*Figure 1*). We recorded the following clinical information for each patient: age, gender, chief complaint, International Neuroblastoma Risk Group Staging System (INRGSS), risk group, primary tumor site, metastatic site, histological subtype, MYCN amplification status, and neuron-specific enolase (NSE) levels within 1 week before imaging (7). The study protocol was approved by the Institutional Review Board of Beijing Friendship Hospital, Capital Medical University (No. 2022-P2-314-01), and the need for individual informed consent for this retrospective study was exempted. This study was conducted in accordance with the Declaration of Helsinki and its subsequent amendments.

Diagnosis, staging, and risk grouping of neuroblastoma

The diagnostic criteria for neuroblastoma (7) are as follows: (I) pathological diagnosis confirmed neuroblastoma via tissue biopsy; (II) identification of characteristic neuroblastoma cells through trephine biopsies, accompanied by significantly elevated levels of catecholamines and their metabolites in blood or urine, and markedly increased serum NSE levels. Neuroblastoma is classified according to the International Neuroblastoma Pathology Classification (8,9), which included: (I) undifferentiated NB; (II) poorly differentiated NB; (III) differentiating NB; (IV) ganglioneuroblastoma (GNB), nodular and intermixed subtype. At the same time, the patients were also staged and categorized into risk groups according to the INRGSS and Children's Oncology Group (COG) guidelines (9).

^{123}I MIBG SPECT/CT image acquisition

All of patients were given 5% Lugol's iodine solution (2 mg/kg children) from 3 days before to 2 days after the imaging to shield the thyroid. In accordance with the clinical status of the patients and the European Association of Nuclear Medicine (EANM)/Society of Nuclear Medicine and Molecular Imaging (SNMMI) guidelines (10), all pediatric participants underwent a standardized medication withdrawal protocol to discontinue drugs known to interfere with MIBG imaging results, including sympathomimetics and tricyclic antidepressants, for at least three half-lives prior to radiotracer administration. The ^{123}I -MIBG SPECT/CT imaging was performed using Siemens Symbia T16 with a low-energy high-resolution collimator. An intravenous injection of 5.2 MBq/kg of ^{123}I -MIBG (provided

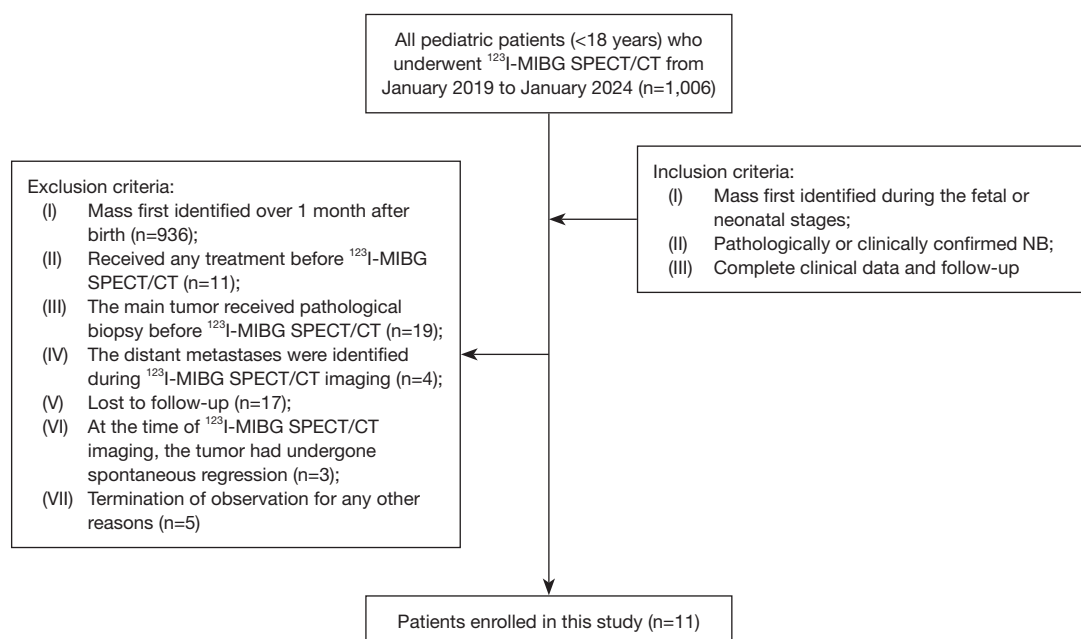


Figure 1 Flowchart of patient inclusion and exclusion. CT, computed tomography; ^{123}I -MIBG, ^{123}I -metaiodobenzylguanidine; NB, neuroblastoma; SPECT, single-photon emission computed tomography.

by Beijing Atomic High-tech Co., Ltd., Beijing, China) was administered 20–24 hours prior to the SPECT/CT scan. First, anteroposterior whole-body planar imaging was performed. Tomographic images were then acquired to characterize lesions that could not be determined through planar imaging and to evaluate suspected lesions. If the planar image was negative, local SPECT/CT tomographic imaging of the surgical area was performed for further assessment. The SPECT image acquisition parameters were as follows: matrix size 256×256, 60 views (10 s/view). A low-dose CT scan with a tube current of 120 mA and a tube voltage 130 kV was subsequently performed for attenuation correction and anatomical localization.

Image analysis

All images were interpreted by two senior nuclear medicine physicians. The volume of interests (VOIs) for primary tumors were manually delineated using 3Dslicer software (version 4.13.0, Boston, Massachusetts, USA). The VOI for muscle was defined as a 1-cm-diameter sphere of the latissimus dorsi muscle at the level of the second lumbar. The VOI for lung was defined as a 1-cm-diameter sphere of right lower lung (superior segment). The VOI for mediastinum blood pool was defined by a 0.3-cm-diameter

sphere within descending aorta. The maximum and mean counts for tumor, muscle, lung and mediastinum blood pool VOIs were recorded. The ratios of the maximum count and the mean count were calculated using the following formula (11,12):

- ❖ $T/M_{\max} = \text{tumor VOI}_{\max} / \text{muscle VOI}_{\max}$;
- ❖ $T/M_{\text{mean}} = \text{tumor VOI}_{\text{mean}} / \text{muscle VOI}_{\text{mean}}$;
- ❖ $T/L_{\max} = \text{tumor VOI}_{\max} / \text{lung VOI}_{\max}$;
- ❖ $T/L_{\text{mean}} = \text{tumor VOI}_{\text{mean}} / \text{lung VOI}_{\text{mean}}$;
- ❖ $T/A_{\max} = \text{TumorVOI}_{\max} / \text{descending aorta VOI}_{\max}$;
- ❖ $T/A_{\text{mean}} = \text{TumorVOI}_{\text{mean}} / \text{descending aorta VOI}_{\text{mean}}$.

Follow-up

If ^{123}I -MIBG SPECT/CT imaging reveals a solitary mass, a mandatory 3-month observation period is instituted. During this period, monthly abdominal ultrasonography and serum NSE level monitoring are conducted, with no therapeutic interventions administered during the observation period. Based on follow-up results, patients were categorized into spontaneous regression group and non-spontaneous regression group. Patients classified as having non-spontaneous regression if one of the following conditions were met: (I) the lesion remains unchanged or

Table 1 The population characteristics of 11 localized congenital neuroblastoma patients

Variables	Categorization	Value
Gender	Boys	7 (63.6)
	Girls	4 (36.4)
Age	Age at identified [range]	38 weeks before birth [31 weeks before birth – 28 days after birth]
	Age at surgery (months) [range]	7 [1–10]
Initial manifestation	Identified during pregnancy	7 (63.6)
	Asymptomatic	1 (9.1)
	Bloated/abdominal masses	2 (18.2)
	Fever	1 (9.1)
Pathological classification	NB (poorly differentiated subtype)	6 (54.5)
	NB (differentiating subtype)	1 (9.1)
	GNB (nodular)	1 (9.1)
	Unclassified	3 (27.3)
Serum NSE (ng/mL) [range]	–	37.0 [11.0–164.0]
INRGSS	L1	8 (72.7)
	L2	1 (9.1)
	L3	2 (18.2)
Risk group	Low risk	6 (54.5)
	Intermediate risk	2 (18.2)
	Unclassified	3 (27.3)

Data are presented as n (%) or median [range]. GNB, ganglioneuroblastoma; INRGSS, International Neuroblastoma Risk Group Staging System; NB, neuroblastoma; NSE, neuron-specific enolase.

increases in size after 3 months; (II) the serum NSE levels continue to rise. Conversely, patients were classified as having spontaneous regression if the following conditions were met: (I) the lesion demonstrates a continuous decreasing trend; (II) the serum NSE levels do not show sustained elevation over 3-month period. Though the lesion demonstrates a continuous decreasing trend or if an emergency arises, appropriate therapeutic interventions should be promptly implemented based on the clinical requirements or the preferences of the guardians, these patients were ultimately excluded from the study population.

Statistical analysis

Statistical analyses were conducted using SPSS 26.0 software. Categorical variables were expressed as counts and percentage [n (%)]. Continuous variables were analyzed

for normality using Shapiro-Wilk test and are reported as median with interquartile ranges. Patients were divided into two groups: the spontaneous regression group, and the non-spontaneous regression group. The Mann-Whitney *U* test was used to compare the continuous variables between two groups. The threshold for statistical significance was set at $P=0.05$.

Results

We retrospectively analyzed data from 11 patients (7 males, 4 females) in this study. The patient characteristics are summarized in *Table 1*. At the time of initial diagnosis, all patients underwent serum NSE testing, with 7 patients exhibiting elevated serum NSE levels. All patients also underwent genetic testing, and none showed MYCN amplification. During the follow-up period (range, 355–1,841 days), all patients survived. Eight patients underwent

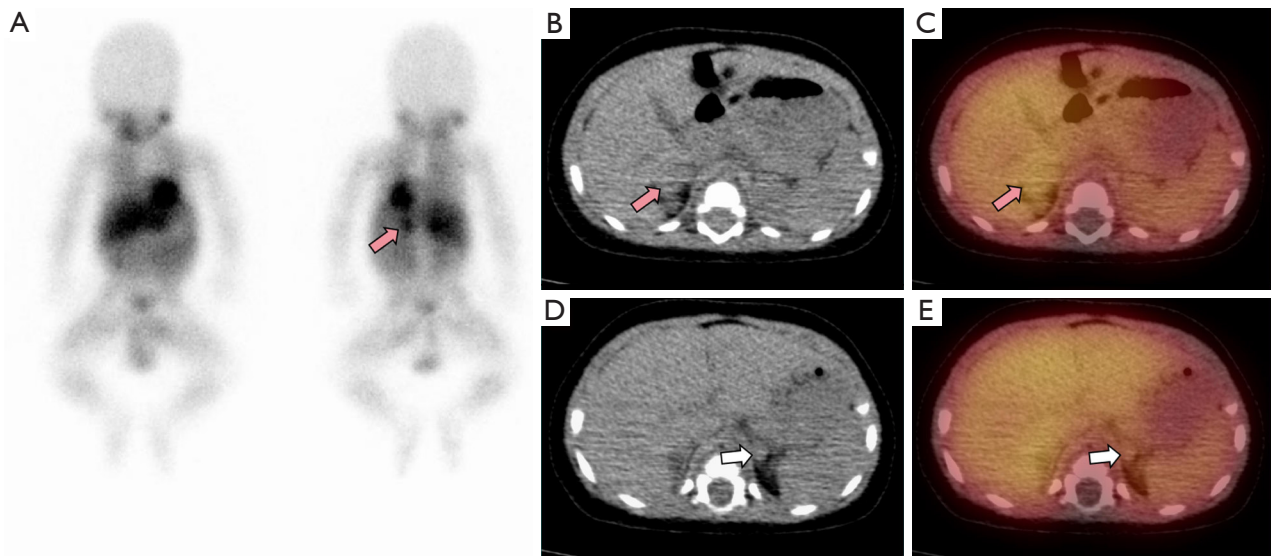


Figure 2 ^{123}I -MIBG SPECT/CT performances in a patient with localized congenital neuroblastoma. A 2-month-old girl (patient No. 7) was identified with bilateral masses on prenatal ultrasound. ^{123}I -MIBG SPECT/CT images revealed masses in the right adrenal gland (A-C, pink arrow) and left adrenal gland (D,E, white arrow). The right mass was larger and exhibited higher uptake of the imaging agent compared to the left mass. During follow-up, both adrenal masses demonstrated a continuous decreasing trend. CT, computed tomography; ^{123}I -MIBG, ^{123}I -metaiodobenzylguanidine; SPECT, single-photon emission computed tomography.

surgery based on clinical indications or parental preferences.

Nine patients presented as solid masses, demonstrating varying degrees of increased uptake of the imaging agent. Two patients exhibited complex cystic and solid masses, with the solid components showing increased uptake of imaging agents. The morphological characteristics of lesions in congenital neuroblastoma exhibit variability. The most common lesion morphology was round or oval-shaped (7 cases), followed by irregular shapes (3 cases) and elongated strip-like forms (1 case). The dimensions of the primary lesions varied, with maximum diameters ranging from 1.3 to 10.9 cm. Notably, three cases featured lesions that extended across the midline, while five cases were positive for imaging-defined risk factors (IDRFs). Most lesions showed calcification foci (7 cases). These calcification foci varied in shape and location, appearing as punctate or large masses and located either within the lesion or at its edge. During follow-up, spontaneous regression was observed in three patients. All three patients had localized solid masses with maximum diameters not exceeding 3.1 cm. None of these lesions crossed the midline, and all were negative for IDRFs. One patient presented as bilateral adrenal masses, with the right mass being larger and exhibiting higher imaging agent uptake than the left (Figure 2). No metastasis

was detected in other sites in any of the three patients. Detailed characteristics of each patient are summarized in Table 2.

The detailed ^{123}I -MIBG SPECT/CT parameters for each patient are summarized in Table 3. The Mann-Whitney *U* test showed significant difference between two groups in terms of maximum diameter, T/Mmax, T/Lmax, T/Amax and serum NSE level (Table 4).

Discussion

This study analyzed imaging features of ^{123}I -MIBG SPECT/CT and clinical characteristics in congenital neuroblastoma patients presenting with solitary mass. Our findings demonstrated that all spontaneously regressed lesions were localized neuroblastoma that did not cross the midline and were negative for IDRFs, and had a maximum diameter of less than 3.1 cm. The uptake of ^{123}I -MIBG uptake by the tumor was significantly associated with the spontaneous regression in localized congenital neuroblastoma.

Since Gadwood *et al.* (13) reported the first case of neuroblastoma detected during pregnancy in 1983, congenital neuroblastoma has gradually become one of the most common malignant solid tumors in neonates.

Table 2 Detailed characteristics of 11 patients

Patient No.	Gender	Primary tumor sites	Maximum diameter (cm)	IDRFs	Complication	Spontaneous regression	Treatment
1	M	Right adrenal gland	4.6	Positive	–	No	Surgery
2	M	Right adrenal gland	2.7	Negative	–	Yes	–
3	M	Right adrenal gland	4.8	Negative	–	No	Surgery
4	M	Right adrenal gland	1.3	Negative	–	Yes	–
5	M	Right adrenal gland	5.1	Negative	–	No	Surgery
6	F	Mediastinum	6.3	Positive	–	No	Surgery
7	F	Retroperitoneal adrenal gland	1.9	Negative	–	Yes	–
8	F	Left adrenal gland	3.1	Positive	–	No	Surgery
9	M	Mediastinum	10.9	Positive	Hydronephrosis	No	Surgery
10	M	Right adrenal gland	4.0	Positive	–	No	Surgery
11	F	Left adrenal gland	5.9	Negative	Umbilical hernia	No	Surgery

F, female; IDRFs, imaging-defined risk factors; M, male.

Table 3 Detailed ^{123}I MIBG SPECT/CT parameters of 11 patients

Patient No.	T/Mmax	T/Mmean	T/Lmax	T/Lmean	T/Amax	T/Amean
1	7.7	2.9	7.7	3.4	13.4	5.5
2	3.1	3.4	3.0	2.7	3.0	2.8
3	5.0	2.7	5.9	3.1	7.8	4.0
4	4.2	5.4	3.1	2.8	3.2	3.0
5	8.5	4.4	6.3	3.3	9.5	4.9
6	16.6	7.0	6.2	2.2	6.8	2.4
7	3.7	4.2	2.4	2.4	2.8	2.9
8	4.5	4.4	2.9	2.8	3.9	3.8
9	22.3	10.8	32.2	15.6	39.5	19.1
10	8.4	5.0	10.2	4.9	10.4	3.9
11	5.3	3.8	3.5	2.0	4.0	2.2

T/Mmax = tumor VOI_{max} /muscle VOI_{max} ; T/Mmean = tumor VOI_{mean} /muscle VOI_{mean} ; T/Lmax = tumor VOI_{max} /lung VOI_{max} ; T/Lmean = tumor VOI_{mean} /lung VOI_{mean} ; T/Amax = tumor VOI_{max} /descending aorta VOI_{max} ; T/Amean = tumor VOI_{mean} /descending aorta VOI_{mean} . CT, computed tomography; ^{123}I -MIBG, ^{123}I -metaiodobenzylguanidine; NSE, neuron-specific enolase; SPECT, single-photon emission computed tomography; VOI, volume of interest.

Approximately 70% of congenital neuroblastoma are identified during the neonatal period (14). However, in our study, the majority of patients (63.6%) were diagnosed with masses during pregnancy.

Surgery is the mainstay treatment for neuroblastoma. However, for patients with localized congenital

neuroblastoma, previous studies have shown that the adverse outcomes associated with surgery can exceed those of the disease itself (15-17), due to the significantly different biological behavior of congenital neuroblastoma compared to neuroblastoma in older patients (18,19). Spontaneous regression is more common in congenital neuroblastoma

Table 4 ¹²³I-MIBG SPECT/CT parameters and Mann-Whitney *U* test

Parameters	Non-spontaneous regression group (n=8)	Spontaneous regression group (n=3)	P
Maximum diameter (cm)	5.0	1.9	0.012
T/Mmax	8.0	3.7	0.012
T/Mmean	4.4	4.2	0.921
T/Lmax	6.3	3.0	0.048
T/Lmean	3.2	2.7	0.279
T/Amax	8.7	3.0	0.012
T/Ameanx	4.0	2.9	0.279
NSE (ng/mL)	40.8	12.2	0.009

T/Mmax = tumor VOI_{max}/muscle VOI_{max}; T/Mmean = tumor VOI_{mean}/muscle VOI_{mean}; T/Lmax = tumor VOI_{max}/lung VOI_{max}; T/Lmean = tumor VOI_{mean}/lung VOI_{mean}; T/Amax = tumor VOI_{max}/descending aorta VOI_{max}; T/Amean = tumor VOI_{mean}/descending aorta VOI_{mean}. CT, computed tomography; ¹²³I-MIBG, ¹²³I-metaiodobenzylguanidine; NSE, neuron-specific enolase; SPECT, single-photon emission computed tomography; VOI, volume of interest.

patients than in older patients (20-22). Several possible explanations for the spontaneous regression include high TrkA expression (3), the absence of nerve growth factor (NGF) (4) in the microenvironment, telomere shortening (5), immune response (23) and epigenetic modifications (24,25). Nuchtern *et al.* (26) concluded that the young infants with the small tumor volume and stable levels of urinary vanillylmandelic acid and homovanillic acid are safe for expectant observation. However, further studies are needed to confirm which biomarkers can predict the spontaneous regression in localized congenital neuroblastoma.

MIBG is a specific imaging agent for neuroblastoma, with a reported sensitivity of 88% to 93% and specificity of 83% to 92% in diagnosing neuroblastoma (26). Prior research (24,27,28) has demonstrated that the semi-quantitative scoring system of ¹²³I-MIBG SPECT/CT holds significant importance in the therapeutic management of neuroblastoma. In our study, we identified three cases of children in the MS stage (metastatic disease in children younger than 18 months, with metastases confined to skin, liver, and/or bone marrow), where ¹²³I-MIBG SPECT/CT played a crucial role in diagnosis and accurate staging (Table S1, Figure S1).

Although previous studies have reported that masses with higher ¹²³I-MIBG uptake are generally associated with poor biological behavior (11,29), the predictive value of the spontaneous regression in localized congenital neuroblastoma remains to be further investigated and demonstrated. In our study, the VOIs for muscle were obtained using a 1-cm-diameter sphere within the latissimus

dorsi muscle at the level of the second lumbar, consistent with previous studies (12,30). Additionally, our study incorporated some new indices as references. The VOIs for lung were obtained using a 1-cm-diameter sphere in the superior segment of right lower lobe, while we attempted to avoid the major pulmonary vessels as much as possible during delineation. However, MIBG uptake by the lung is related to pulmonary vascular endothelial function (31), and all patients underwent free-breath SPECT/CT, which may introduce deviations in the count results. Previous studies (11,29,30,32) used the liver as a reference organ; however, the liver exhibits physiological uptake, and prior study has demonstrated that MIBG uptake in the liver can vary (33), potentially leading to deviations in the count results. Although our result indicated significant difference between two groups in terms of T/Mmax, T/Lmax, T/Amax. We believe that the tumor-to muscle ratios would be more appropriate for indicating the degree of MIBG uptake by the primary tumor.

Additionally, our findings indicate that serum NSE levels exhibited a significant difference between two groups. Previous studies (34,35) have demonstrated that pre-treatment serum NSE levels were useful for guiding treatment decisions. However, further research is needed to determine an optimal cutoff value.

There are some limitations in this study. First, it is a retrospective single-center study with a relatively small sample size, which may introduce bias in patient selection and parameter inclusion, such as lactate dehydrogenase, vanillylmandelic acid and homovanillic acid. Second, the

follow-up duration is relatively short; therefore, further continuous follow-up will be necessary in future research to provide additional support for the verifying the related findings. Moving forward, we plan to conduct a prospective, large-sample, multicenter study incorporating additional relevant predictive variables to address these limitations.

Conclusions

^{123}I -MIBG SPECT/CT is valuable for the diagnosis, staging and monitoring of treatment response in congenital neuroblastoma. The ^{123}I -MIBG uptake of the tumor has been shown to predict the spontaneous regression in localized congenital neuroblastoma.

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None.

Footnote

Reporting Checklist: The authors have completed the STROBE reporting checklist. Available at <https://qims.amegroups.com/article/view/10.21037/qims-2025-524/rc>

Data Sharing Statement: Available at <https://qims.amegroups.com/article/view/10.21037/qims-2025-524/dss>

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Conflicts of Interest: All authors have completed the ICMJE uniform disclosure form (available at <https://qims.amegroups.com/article/view/10.21037/qims-2025-524/coif>). The authors have no conflicts of interest to declare.

Ethical Statement: The authors are accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. The study protocol was approved by the Institutional Review Board of Beijing Friendship Hospital, Capital Medical University (No. 2022-P2-314-01), and the need for individual informed consent for this retrospective study was exempted. This study was conducted in accordance with the Declaration of Helsinki and its subsequent amendments.

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